MAS508-G Control Theory
Lecture 3: State Space Design

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L3. State Space Design

a) Time Response
b) Controllability
c) Observability
d) Pole Placement Control
e) Observer Design
f) Output Feedback Control
g) Optimal Control - Linear Quadratic Control (LQR)
h) Optimal Control – LQG

Textbook, “Control Systems Engineering”, Chapters 4 &12

L3c. Observability

Book “Control Systems Engineering”, Chapter 12.6

What is observability?

A system is *observable* if the states of the system can be determined from its measured outputs.

What determines observability?

The number, type, and/or configuration of sensors on the system

Why do we need it?

- Many systems **do not have a sensor measuring every state** or element in the system
- But if a system is *observable*, then you can *estimate* the states of the system
- If you know all the states, then you can design a control system

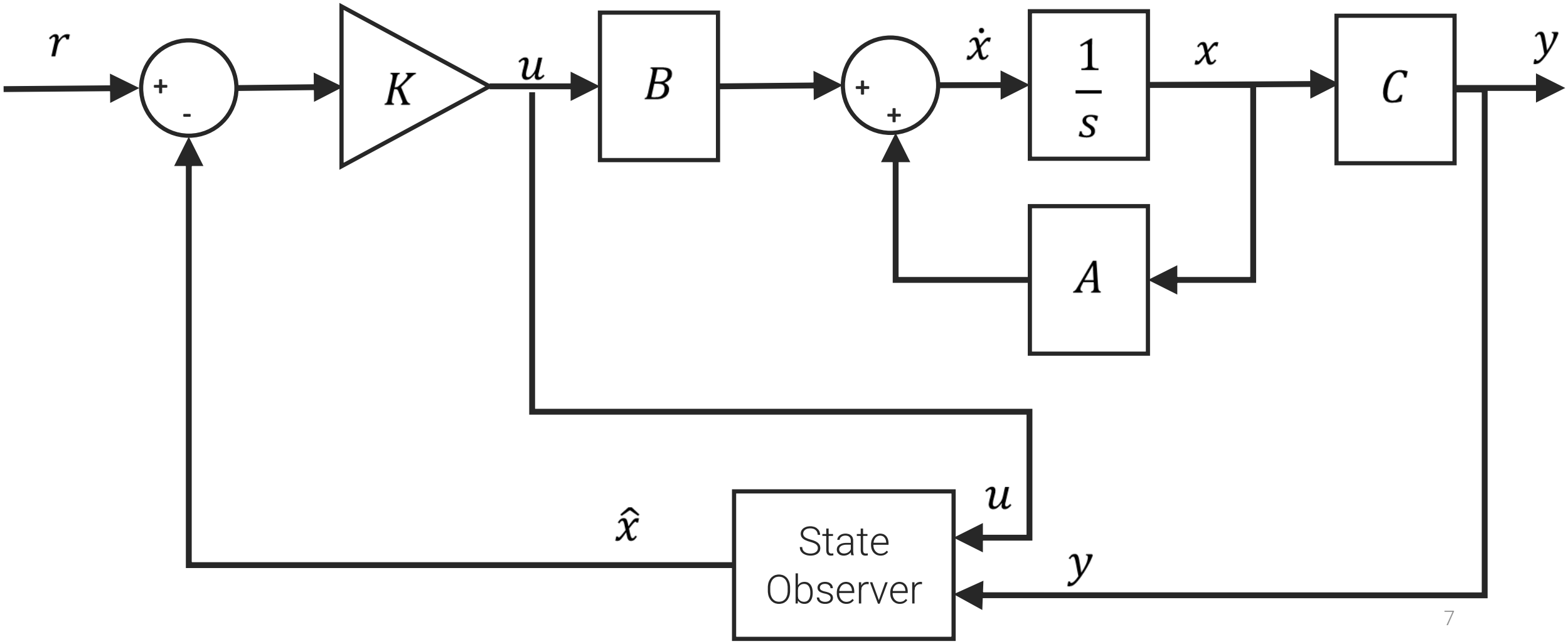


Quad-rotor drones are a prime example.
The on-board Inertial Module Unit (IMU) sensor (e.g. accelerometer, gyroscope) can be used to assess the pose of the vehicle.

Observer

- Not all the states in the system may be directly measurable.
- In this situation, you need to *estimate* the states using the measured outputs.
- This is known as an **estimator**, or an **observer**.

Observer



Definition

- A system is *observable* if the initial state $\boldsymbol{x(t_0)}$ can be determined from the system output $\boldsymbol{y(t)}$ and the control input $\boldsymbol{u(t)}$ in finite time.

- Observability is the ability to deduce the state variables from a knowledge of the input $u(t)$ and the output $y(t)$.

- **Observability** is a measure of how well internal states of a system can be inferred from knowledge of its external outputs.

Examples

Example: Observable

$$\dot{x} = \begin{bmatrix} 2 & -1 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} x$$

Example: unobservable

$$\dot{x} = \begin{bmatrix} 2 & -1 \\ 0 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 0 & 1 \end{bmatrix} x$$

Example: unobservable

$$\dot{x} = \begin{bmatrix} 2 & -1 \\ -5 & -2 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$
$$y = \begin{bmatrix} 1 & 1 \end{bmatrix} x$$

Observability Matrix

State-space model of linear system:

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

where:

A = system matrix $(n \times n)$
B = input matrix $(n \times r)$
C = output matrix $(m \times n)$
D = feedforward matrix $(m \times r)$

The system state

$$x = [x_1 \quad x_2 \quad x_3 \ldots x_{n-1} \quad x_n]^T$$

and control input u.

The observability matrix is defined as:

$$O_{ac} = \begin{bmatrix} C \\ CA \\ CA^2 \\ \vdots \\ CA^{n-2} \\ CA^{n-1} \end{bmatrix}$$

The system is observable **if and only if** the observability matrix is nonsingular.

Matlab command: observability matrix
obsv(A,C)

Testing for Observability

Rank Test:
The observability matrix is nonsingular if and only if its rank equals the number of system states.

$$rank(O_{ac}) = n$$

where n = # of states in the system.

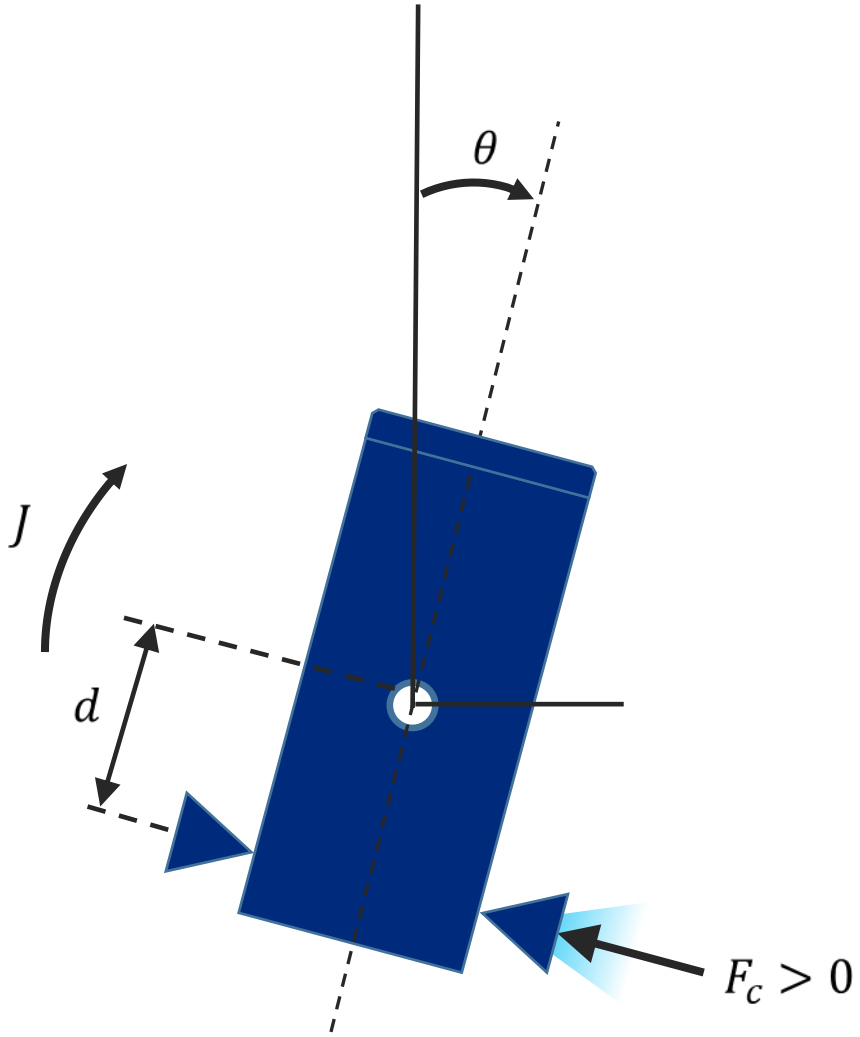
Determinant Test:
The observability matrix is nonsingular if and only if its determinant is nonzero.

$$|O_{ac}| \neq 0$$

You can use either of these two methods to check if the system is observable.

Determinant Test: only available for square matrix.

Example 1: Satellite Heading Control



Source: Courtesy Thaicom PLC and Space Systems/Loral

$$\dot{\theta} = \omega$$

$$\dot{\omega} = \frac{d}{J} F_c$$

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ d/J \end{bmatrix} u$$

$$y = [0 \quad 1]x$$

Only measuring angular velocity

$$O_{ac} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Can you design an observer when measuring only angular velocity?

Rank test:

$$rank(O_{ac}) = 1 < 2$$

It's not observable. To make it observable you would need a position sensor.

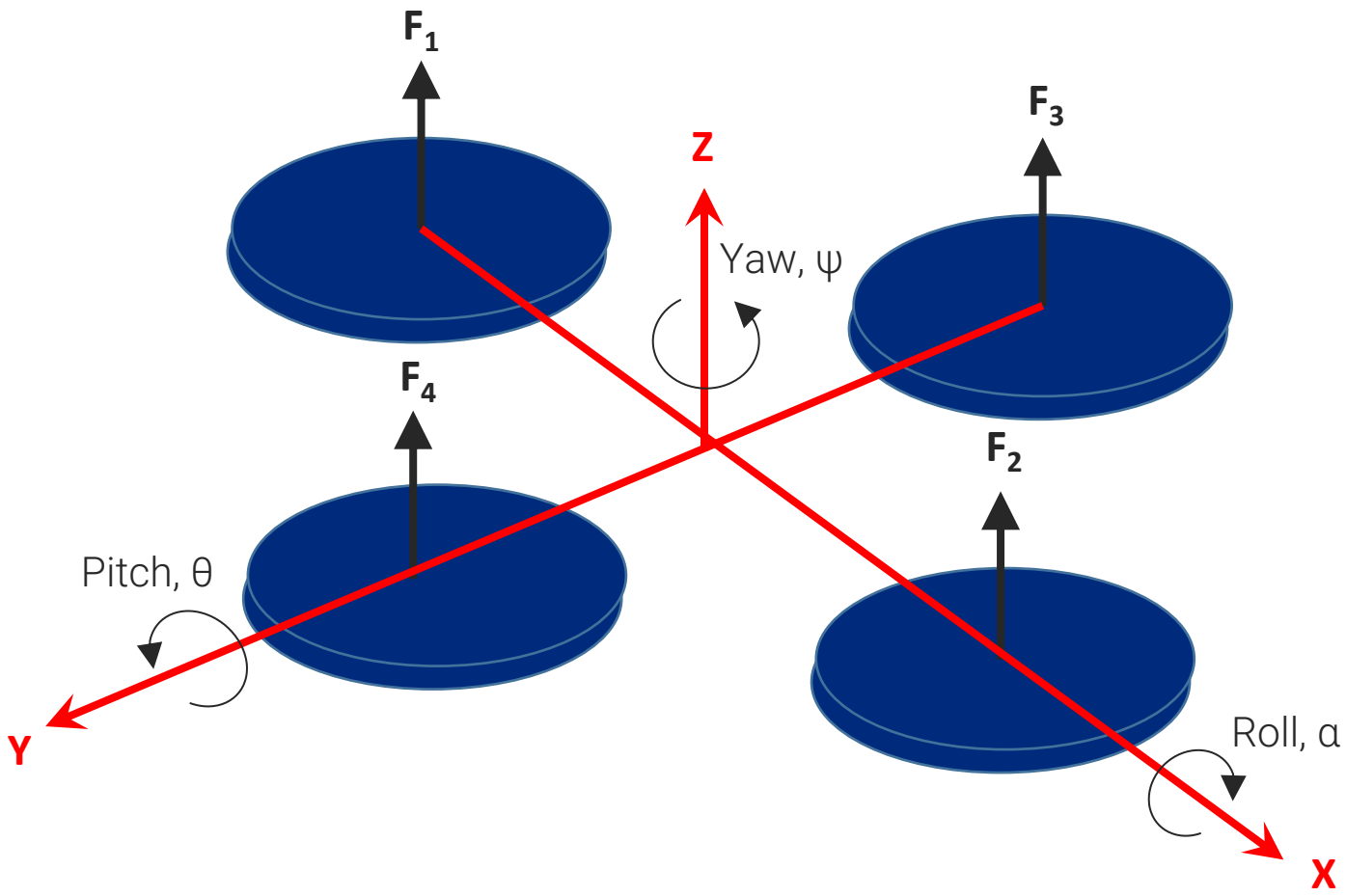
Autonomous Aerial Vehicles (UAVs)

- Position/pose of aircrafts cannot be measured directly and precisely enough for control
- Pose of the aerial vehicle (pitch, roll, yaw) relative to the body frame can be *estimated* using the IMU measurements



Pose can be estimated using a 9 DOF IMU (3 DOF gyroscope, accelerometer, and magnetometer)

Quadrotor Drone



The pitch, roll, and yaw position and velocity are the angular states of the drone:

$$x^T = [\theta \quad \dot{\theta} \quad \alpha \quad \dot{\alpha} \quad \psi \quad \dot{\psi}]$$

IMU measures the angular velocity and acceleration of those (not the position directly):

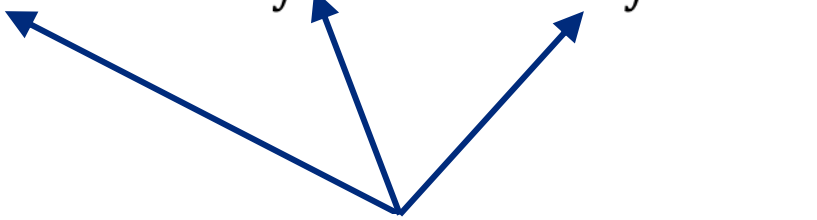
$$y^T = [\dot{\theta} \quad \ddot{\theta} \quad \dot{\alpha} \quad \ddot{\alpha} \quad \psi \quad \dot{\psi} \quad \ddot{\psi}]$$

Design an observer that uses all the measured variables in order to obtain an accurate pose estimate

Do I need an observer?

Why can't I just integrate the velocities to get the position?

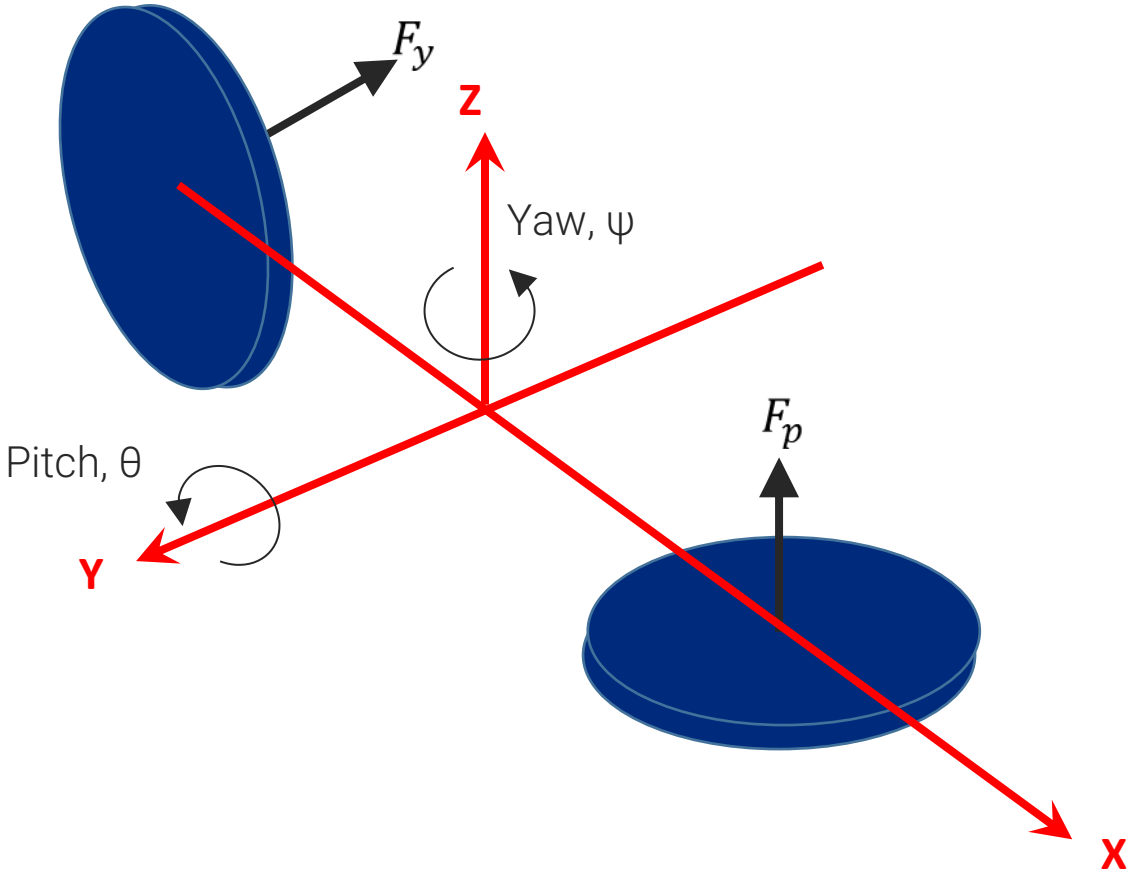
$$\theta = \int \dot{\theta}(t)dt \quad \alpha = \int \dot{\alpha}(t)dt \quad \psi = \int \dot{\psi}(t)dt$$



Integrate measured angular velocities to get position

The issue with integration is your position will drift over time.

Example 2: 2 DOF Helicopter



The pitch and yaw position and velocity are the states of the system:

$$x^T = [\theta \quad \psi \quad \dot{\theta} \quad \dot{\psi}]$$

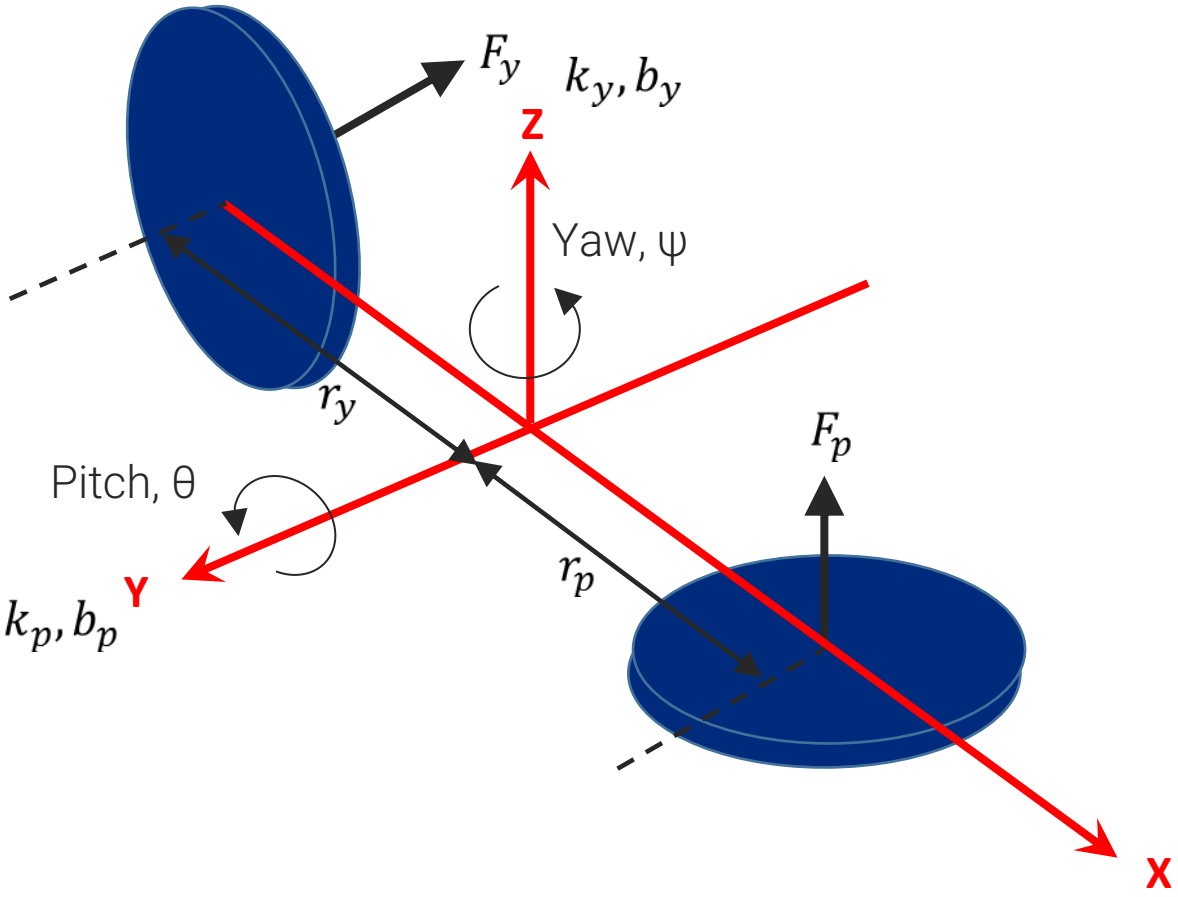
Assume you only have an accelerometer measuring the pitch and yaw acceleration :

$$y^T = [\ddot{\theta} \quad \ddot{\psi}]$$

Note: not measuring position directly

Can you design an **observer** to estimate state x?

Example 2: Model



Simplified linear equations of motions for pitch and yaw axes:

$$J_p\ddot{\theta} + b_p\dot{\theta} + k_p\theta = \tau_p \qquad \tau_p = r_p F_p$$
$$J_y\ddot{\psi} + b_y\dot{\psi} + k_y\psi = \tau_y \qquad \tau_p = r_y F_y$$

The state-space representation of system

$$\dot{\mathrm{x}} = \mathrm{A}x + \mathrm{B}u$$
$$\mathrm{y} = \mathrm{C}x + \mathrm{D}u$$

Given the control input is the torque $u = [\tau_\mathrm{p} \ \ \tau_\mathrm{y}]$ is

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k_p/J_p & 0 & -b_p/J_p & 0 \\ 0 & -k_y/J_y & 0 & -b_y/J_y \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1/Jp & 0 \\ 0 & 1/Jy \end{bmatrix}$$

$$C = \begin{bmatrix} -k_p/J_p & 0 & -b_p/J_p & 0 \\ 0 & -k_y/J_y & 0 & -b_y/J_y \end{bmatrix} \quad D = \begin{bmatrix} 1/Jp & 0 \\ 0 & 1/Jy \end{bmatrix}$$

Example 2: System Parameters

Parameters for pitch θ (rad):

- Inertia of pitch Jp = 0.0217 (kg-m2)
- Damping constant along pitch axis bp = 0.0071 (V-s/rad)
- Friction stiffness along pitch axis Kp = 0.0371 (N-m/rad)

Parameters for yaw ψ (rad):

- Inertial of yaw Jy = 0.0217 (kg-m2)
- Damping constant along yaw axis by = 0.0217 (V-s/rad)
- Friction stiffness along pitch axis Ky = 0.0371 (N-m/rad)

Example 2: Test Observability

Given set of helicopter parameters, the observability matrix is:

$$A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -k_p/J_p & 0 & -b_p/J_p & 0 \\ 0 & -k_y/J_y & 0 & -b_y/J_y \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1.71 & 0 & -0.325 & 0 \\ 0 & -1.71 & 0 & -1.00 \end{bmatrix}$$

$$C = \begin{bmatrix} -k_p/J_p & 0 & -b_p/J_p & 0 \\ 0 & -k_y/J_y & 0 & -b_y/J_y \end{bmatrix} = \begin{bmatrix} -1.71 & 0 & -0.325 & 0 \\ 0 & -1.71 & 0 & -1.00 \end{bmatrix}$$

Measuring pitch and yaw accelerations

$$y^T = [\ddot{\theta} \quad \ddot{\psi}]$$

Observability matrix:

$$O_{ac} = \begin{bmatrix} C \\ CA \\ CA^2 \\ CA^3 \end{bmatrix}$$

Evaluate

$$O_{ac} = \begin{bmatrix} -1.71 & 0 & -0.325 & 0 \\ 0 & -1.71 & 0 & -1.00 \\ 0.556 & 0 & -1.61 & 0 \\ 0 & 1.70 & 0 & -0.703 \\ 2.75 & 0 & 1.08 & 0 \\ 0 & 1.20 & 0 & 2.41 \\ -1.85 & 0 & 2.40 & 0 \\ 0 & -4.10 & 0 & -1.21 \end{bmatrix}$$

Rank test: $rank(O_{ac}) = 4$

It's observable. Therefore, we can design an observer to estimate its pitch and yaw positions.

Example 2: What if you can only measure rates?

Case 1: Measuring pitch and yaw rates

$$y = \begin{bmatrix} \dot{\theta} \\ \dot{\psi} \end{bmatrix} \qquad C = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$O_{ac} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1.71 & 0 & -0.325 & 0 \\ 0 & -1.71 & 0 & -1.00 \\ 0.556 & 0 & -1.61 & 0 \\ 0 & 1.71 & 0 & -0.703 \\ 2.75 & 0 & 1.08 & 0 \\ 0 & 1.20 & 0 & 2.41 \end{bmatrix}$$

Rank test:

$$rank(O_{ac}) = 4$$

It's **observable**.

Case 2: Measuring only yaw rate

$$y = \dot{\psi} \qquad C = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$O_{ac} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & -1.71 & 0 & -1.00 \\ 0 & 0 & -0 & 0 \\ 0 & 1.71 & 0 & -0.703 \\ 0 & 0 & 0 & 0 \\ 0 & 1.20 & 0 & 2.41 \end{bmatrix}$$

Rank test:

$$rank(O_{ac}) = 2 < 4$$

It's **not observable**.

Takeaways

Benefits

- If it's observable, then you can design an observer to estimate the states
- Can determine if a given system has enough or the correct type of sensors

Drawbacks

- Computing observability matrix can be tedious
- Need system dynamic model

State-Space Models

- Controllable Canonical Form
- Observable Canonical Form

Controllable Canonical Form

The controllable canonical form is the form where the characteristic polynomial of the system appears explicitly **in the last row of the A matrix**.

$$y^{(n)} + a_{n-1}\,y^{(n-1)} + \cdots + a_1\dot{y} + a_0 y = b_{n-1}u^{(n-1)} + b_{n-2}\,u^{(n-2)} + \cdots + b_1\dot{u} + b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

Define

$x_1 = z, x_2 = \dot{z}, x_3 = \ddot{z}\,, x_{n-1} = z^{(n-2)}\,, x_n = z^{(n-1)}$

$x = [x_1\ x_2\ \ldots x_n]^T$

$y = [b_0\ \ b_1\ \ldots.\ b_{n-2}\ \ b_{n-1}]\,x$

$$\dot{x} = A_c x + B_c u$$
$$y = C_c x + D_c u$$

$$A_c = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & & 0 \\ & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}$$

$$B_c = [0\ \ 0\ \cdots 0\ \ 1]^T$$

$$C_c = [b_0\ \ b_1\ \ldots.\ b_{n-2}\ \ b_{n-1}]$$

$$D_c = 0$$

Controllable Canonical Form

- The controllable canonical form is the form where the characteristic polynomial of the system appears explicitly in the last row of the A matrix.
- This state-space realization is called controllable canonical form because the resulting model is guaranteed to be controllable.
- The controllable canonical form is useful for controller design using pole placement method.
- However, the transformation of the system to companion form is based on the controllability matrix which is almost always numerically singular for mid-range orders. Hence, avoid using the controllable form for computation when possible.

Observable Canonical Form

$$y^{(n)} + a_{n-1}\,y^{(n-1)} + \cdots + a_1\dot{y} + a_0 y = b_{n-1}u^{(n-1)} + b_{n-2}\,u^{(n-2)} + \cdots + b_1\dot{u} + b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

$$(s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0)Y(s) = (b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_1 s + b_0)\mathrm{U}(s)$$

$$s^n Y(s) = (b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots + b_1 s + b_0)\mathrm{U}(s) - (a_{n-1}s^{n-1} + a_{n-2}s^{n-2} \ldots + a_1 s + a_0)Y(s)$$

$$sY(s) = \left(b_{n-1} + \frac{b_{n-2}}{s} + \cdots + \frac{b_1}{s^{n-2}} + \frac{b_0}{s^{n-1}}\right)\mathrm{U}(s) - \left(a_{n-1} + \frac{a_{n-2}}{s} \ldots + \frac{a_1}{s^{n-2}} + \frac{a_0}{s^{n-1}}\right)Y(s)$$

$$z_1(s) = Y(s)$$

$$z_2(s) = \left(\frac{b_{n-2}}{s} + \cdots + \frac{b_1}{s^{n-2}} + \frac{b_0}{s^{n-1}}\right)\mathrm{U}(s) - \left(\frac{a_{n-2}}{s} + \cdots + \frac{a_1}{s^{n-2}} + \frac{a_0}{s^{n-1}}\right)Y(s)$$

$$z_3(s) = \left(\frac{b_{n-3}}{s} + \cdots + \frac{b_1}{s^{n-3}} + \frac{b_0}{s^{n-2}}\right)\mathrm{U}(s) - \left(\frac{a_{n-3}}{s} + \cdots + + \frac{a_1}{s^{n-3}} + \frac{a_0}{s^{n-2}}\right)Y(s)$$

$$\vdots$$

$$z_n(s) = \left(\frac{b_0}{s}\right)\mathrm{U}(s) - \left(\frac{a_0}{s}\right)Y(s)$$

Observable Canonical Form

The observable canonical form is the form where the characteristic polynomial of the system appears explicitly in the rightmost column of the A matrix.

$$y^{(n)} + a_{n-1}\,y^{(n-1)} + \cdots + a_1\dot{y} + a_0 y = b_{n-1}u^{(n-1)} + b_{n-2}\,u^{(n-2)} + \cdots + b_1\dot{u} + b_0 u$$

$$\frac{Y(s)}{U(s)} = \frac{b_{n-1}s^{n-1} + b_{n-2}s^{n-2} + \cdots b_1 s + b_0}{s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0}$$

Define

$$x = [x_1\ x_2\ \ldots x_n]^T$$

$$\dot{x} = A_o x + B_o u$$
$$y = C_o x + D_o u$$

$$A_o = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 & -a_0 \\ 1 & 0 & 0 & \cdots & 0 & -a_1 \\ 0 & 1 & 0 & \cdots & 0 & -a_2 \\ 0 & 0 & 1 & \cdots & 0 & -a_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & -a_{n-1} \end{bmatrix}$$

$$B_o = [b_0\ \ b_1\ \ \ldots.\ b_{n-2}\ \ b_{n-1}]^T$$

$$C_o = [0\ \ 0\ \cdots 0\ \ 1]$$

$$D_o = 0$$

Observable Canonical Form

- The observable canonical form is the form where the characteristic polynomial of the system appears explicitly in the rightmost column of the A matrix.

- This state-space realization is called observable canonical form because the resulting model is guaranteed to be observable.

- The observable canonical form is useful for observer design using pole placement method.

The relationship between the observable and controllable canonical forms

- The relationship between the observable and controllable canonical realizations are as follows:

$$A_c = A_o^T$$
$$B_c = C_o^T$$
$$C_c = B_o^T$$
$$D_c = D_o^T$$